



WEB TECHNOLOGY

HTML5



Course Outline

Lecture 1 :

1.1 Introduction to Internet,

1.2 World Wide Web and

1.3 Markup Language: Popular Usage of the Web

1-1 INTRODUCTION TO INTERNET

The Internet

The **largest network** in the world that **interconnects** millions of individual networks all over the world.

Academic, business, social and government networks are linked from **local to global** scope.

Popular terms: “**Network of Networks**”, “**Information Highway**”.



The Internet

- ✓ No one owns it!
- ✓ It was originally developed by department of defense in United State in 1969.
- ✓ To access, an existing network need to pay a small registration fee and agree to certain standards based on the **TCP/IP**.



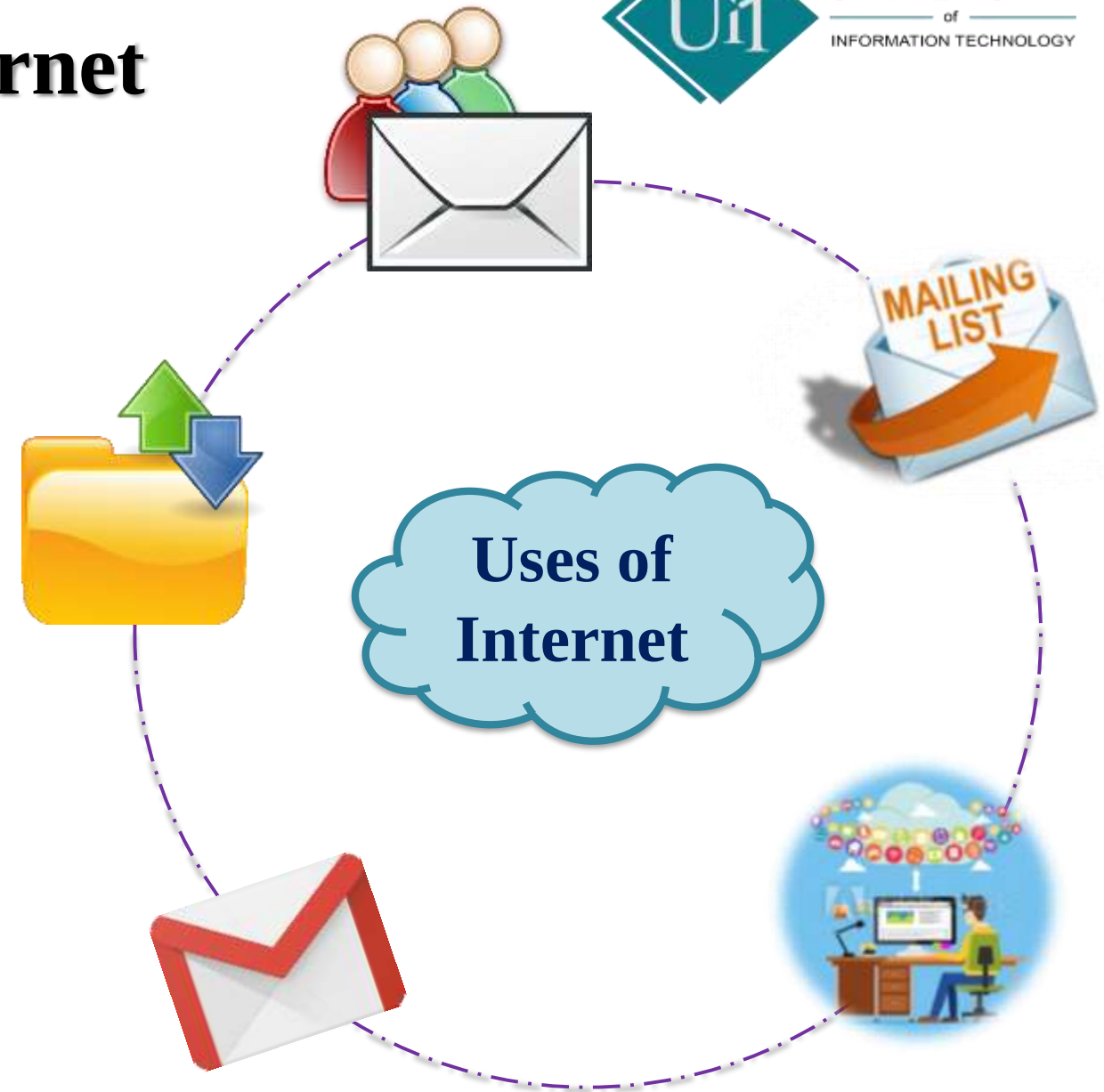
Transmission Control Protocol/
Internet Protocol

The uses of the Internet



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- ✓ Send e-mail messages.
- ✓ Send (upload) or receive (download) files between computers.
- ✓ Participate in discussion groups, such as mailing lists and newsgroups.
- ✓ Surfing the **web**.



The uses of the Internet (cont'd)

✓ Send e-mail messages.



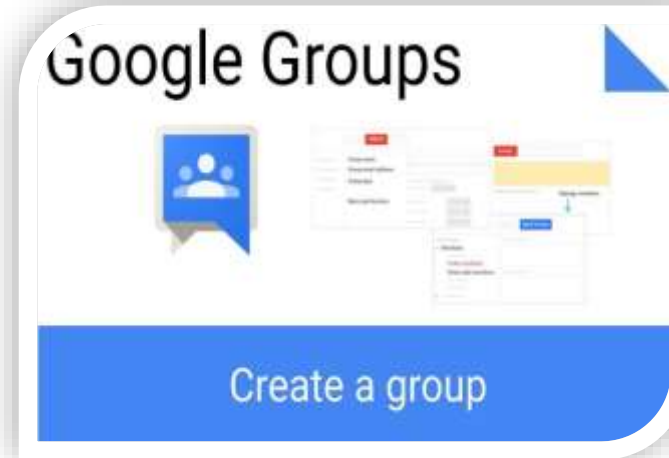
The uses of the Internet (cont'd)

- ✓ Send (upload) or receive (download) files between computers.



The uses of the Internet (cont'd)

- ✓ Participate in discussion groups, such as mailing lists, newsgroups.



The uses of the Internet (cont'd)

✓ Surfing the Web.



How to access the Internet?

- ✓ Direct access to the Internet using special high-speed *communication lines and equipment*.
- ✓ Access through the organization's *local area networks (LAN)* or through their own *personal computers*.
- ✓ Access the Internet through *Internet Service Provider (ISP)*.

1-2 THE WORLD WIDE WEB (WWW)

What is Web?



Sir Tim Berners-Lee, a British computer scientist, invented the World Wide Web in 1989.



The **Web (World Wide Web)** is an information system identified by Uniform Resource Locators: URLs (<https://www.example.com/>) of the Internet

What is Web? (cont'd)



The Web consists of information organized into **Web pages** containing text and graphic images.



The **Web** also contains *hypertext links*, or *highlighted keywords and images* that lead to related information.



What is Website?



A collection of linked Web pages that has a *common theme or focus* is called a **Website**.



The main page that all of the pages on a particular Website are organized around and link back to is called the site's **home page**.



How to access the Web?

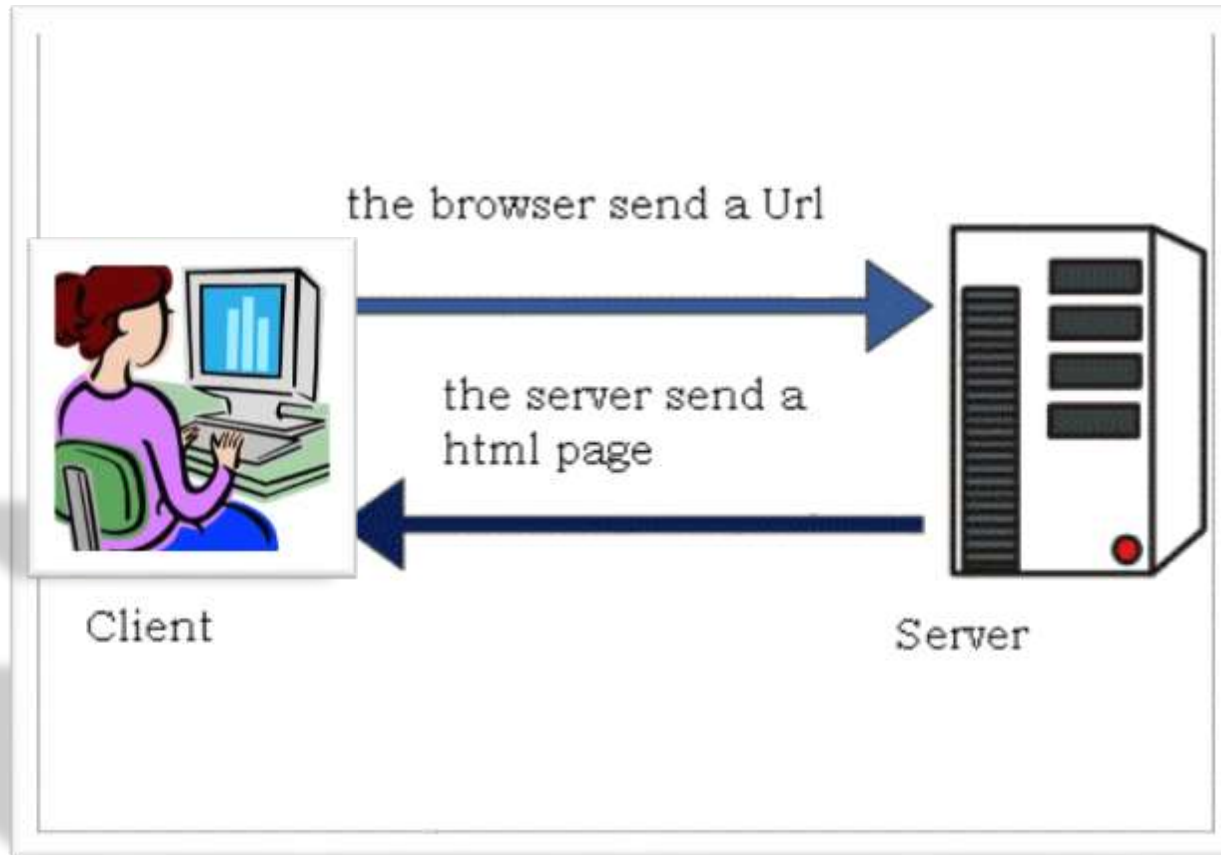
- ✓ If there has **Internet connection**, a special software called a **browser** is needed to access the Web.
- ✓ Web browsers are **specialized programs** used to connect remote computers, open and transfer files, display text and images.
- ✓ Examples of Web browser: Internet Explorer, Firefox, Chrome, Opera, Safari, etc



Client/Server Structure of the Web

- ✓ A **web server** is *server software*, or *hardware dedicated to running this software*, that can satisfy client requests on the World Wide Web.
- ✓ When you use your Internet connection to become part of the Web, your computer becomes a **Web client** in a worldwide client/server network.

Client/Server Structure of the Web



Quiz (1)

A collection of linked Web pages that has a common theme or focus is called ...?

- A. Home page
- B. Website
- C. Link



Quiz (2)

When I use my Internet connection, my computer becomes a Web ... in a worldwide client/server network?

- A. browser
- B. server
- C. client



1-3 MARKUP LANGUAGE: POPULAR USES OF THE WEB

Hypertext Markup Language (HTML)

The public files on the web servers are **ordinary text files**, much like the files used by word-processing software.

To allow Web browser software to read them, **the text must be formatted** according to a generally accepted standard.

The standard used on the web is *HyperText Markup Language (HTML)*.

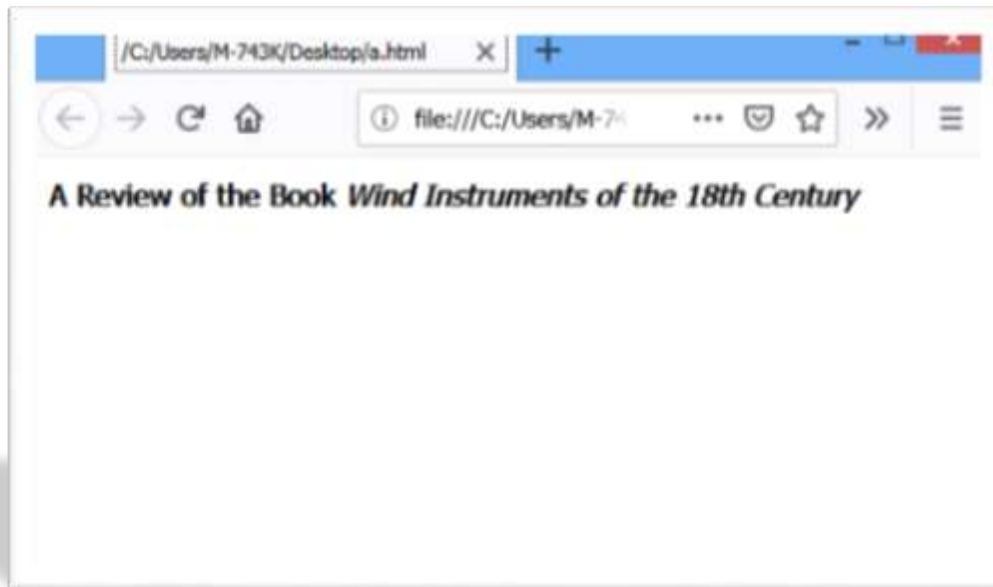
Hypertext Markup Language (HTML) (cont'd)

- ✓ HTML was invented by Sir Tim Berners-Lee in late 1991.
- ✓ HTML uses **codes**, or **tags** `< >`, `</ >`, to tell the web browser software how to display the text contained in the document.
 - For example, a Web browser reading the following line of text:

`<B> A Review of the Book <I> Wind Instruments of the 18th Century</I></B>`

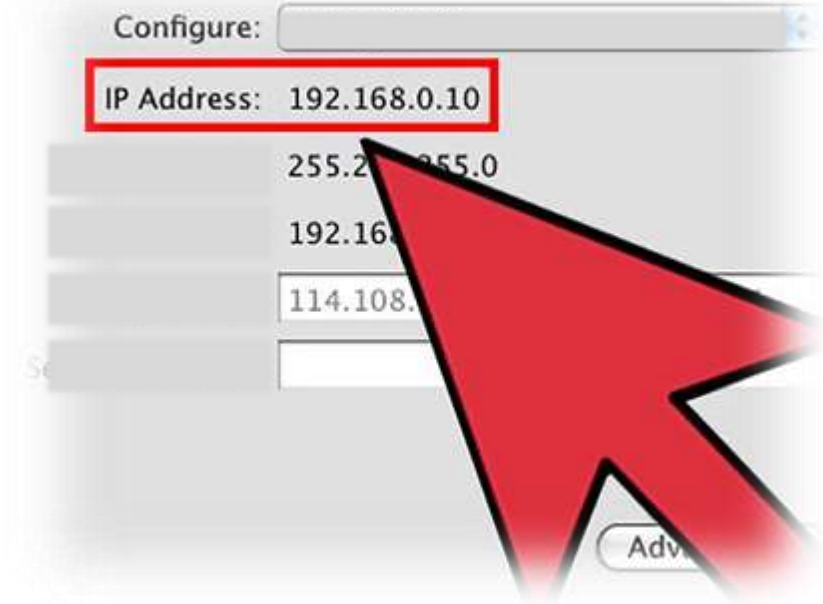
Hypertext Markup Language (HTML) (cont'd)

- the `<B>` and `</B>` tags as instructions to display the entire line of text in **bold** and
- the `<I>` and `</I>` tags as instructions to display the text enclosed by those tags in *italics*.



IP Addressing

- ✓ Each computer on the internet does have a **unique identification number**, called an **IP (Internet Protocol) address**.
- ✓ The IP addressing system currently in use on the Internet uses a **four-part number**.
 - Each part of the address is a **number ranging from 0 to 255**, and each part is separated from the previous part by period,
 - For example, 192.168.0.10



Domain Name Addressing

- ✓ Most web browsers do not use the *IP address* to locate Web sites and individual pages.
- ✓ They use **Domain Name Addressing**.

Domain Name Addressing (cont'd)

- A **domain name** is a **unique name** associated with a specific **IP address** that defines a realm of administrative autonomy, authority or control within the Internet.
- The **program**, which coordinates the IP addresses and domain names for all computers attached to it, is called **DNS (Domain Name System) software**.
- The **host computer** that runs this software is called a **domain name server**.

Domain Name Addressing (cont'd)

- Domain names can include **any number of parts separated by periods**, however most domain names currently in use have **only three or four parts**.
- Domain names follow **hierarchical model** that you can follow from top to bottom if you read the name **from the right to the left**.

Domain Name Addressing (cont'd)

- ✓ For example, the domain name **gsb.uchicago.edu** is
 - the computer connected to the Internet at the **Graduate School of Business (gsb)**, which is an academic unit of the **University of Chicago (uchicago)**, which is an **educational institution (edu)**.
 - **No other computer** on the Internet has the **same domain name!**

Uniform Resource Locators (URL)

- ✓ The IP address and the domain name each identify on the Internet.
- ✓ However, they do not indicate where a Web page's HTML document resides on that computer on the Internet.
- ✓ To identify a Web pages exact location, Web browsers rely on **Uniform Resource Locator (URL)**.
- ✓ A URL is a complete web address used to find a particular web page.

Structure of a Uniform Resource Locators

URL is a *four-part addressing scheme* that tells the Web browser:

1. What **transfer protocol** to use for transporting the file
2. The **domain name** of the computer on which the file resides
3. The **pathname of the folder or directory** on the computer on which the file resides
4. The **name of the file**



`http` => Hypertext Transfer Protocol

How to find information on the Web?

- ✓ A number of **search tools** have been developed and available to you on certain Web sites that provide search services to help you find information.
- ✓ Examples:
 - MSN WebSearch: www.search.msn.com
 - Yahoo: www.yahoo.com
 - Google: www.google.com
 - Bing: www.bing.com

How to find information on the Web? (cont'd)

- ✓ You can find information by **two basic means**.
 1. Search by **Topic** and
 2. Search by **keywords**.
- ✓ Some search services offer **both methods**, others **only one**.
- ✓ For example, Yahoo offers **both**.

Quiz (1)

Which is an example of an IP address?

- A. www.uit.edu.mm
- B. 192.168.240.10
- C. (+95) -1-9664254



Quiz (2)

Which is an example of protocol?

- A. http
- B. google
- C. com



Summary

1.1 Introduction to Internet

- ✓ Internet
- ✓ The uses of Internet
- ✓ How to access the Internet?

1.2 World Wide Web

- ✓ What is Web?
- ✓ What is Website?
- ✓ How to access the Web?
- ✓ Client/Server Structure of the Web



1.3 Markup Language: Popular Usage of the Web

- ✓ Hypertext Markup Language (HTML)
- ✓ IP Addressing, Domain Name Addressing
- ✓ Uniform Resource Locators (URL) and its Structure
- ✓ How to find information on the Web?